git log/history

References

- Visualizing Code: Polyglot Notebooks Repository (YouTube)
- gitstractor (GitHub)
- Neo4j Python Driver

```
Command line execution (CLI mode): Yes
```

Git History - Directory Commit Statistics

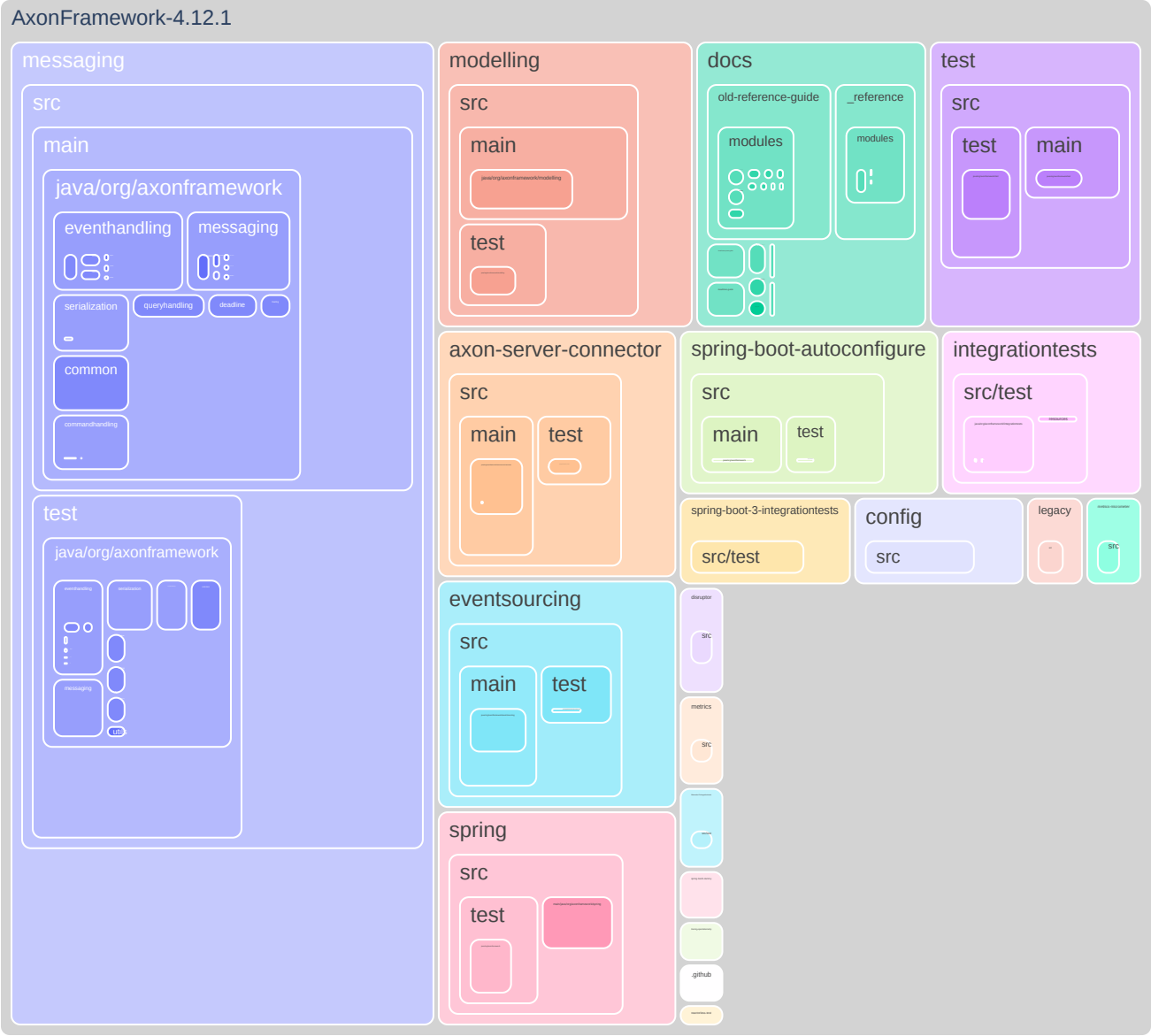
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	564.000000	564.000000	564.000000	564.000000	564.000000	564.000000
mean	24.898936	19.664894	288.570922	310.186170	1187.539007	404.822695
std	120.739280	21.297173	639.823202	170.624646	980.407058	392.535607
min	1.000000	2.000000	3.000000	32.000000	46.000000	32.000000
25%	1.000000	7.000000	27.000000	172.000000	446.000000	263.000000
50%	4.000000	12.000000	82.500000	405.000000	965.000000	405.000000
75%	10.000000	25.000000	270.250000	405.000000	2012.000000	405.000000
max	2294.000000	213.000000	9815.000000	2555.000000	5683.000000	3034.000000

	directoryPath	directoryParentPat
0	AxonFramework-4.12.1/assembly	AxonFramework-4.12.
1	AxonFramework-4.12.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source	AxonFramework-4.12.1/axon-serve connector/src/main/java/org/axonframework/axonserver/connector/heartbea
2	AxonFramework-4.12.1/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-4.12.1/axon-server-connector/src/main/resource
3	AxonFramework-4.12.1/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util	AxonFramework-4.12.1/axon-serve connector/src/test/java/org/axonframework/axonserver/connector/ever
4	AxonFramework-4.12.1/config/src/test/resources/META-INF	AxonFramework-4.12.1/config/src/test/resource
5	AxonFramework-4.12.1/coverage-report	AxonFramework-4.12.
6	AxonFramework-4.12.1/docs/_playbook/localLinks	AxonFramework-4.12.1/docs/_playboo
7	AxonFramework-4.12.1/docs/_reference/modules/ROOT/attachments	AxonFramework-4.12.1/docs/_reference/modules/ROO
8	AxonFramework-4.12.1/docs/_reference/modules/ROOT/partials	AxonFramework-4.12.1/docs/_reference/modules/ROO
9	AxonFramework-4.12.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.12.1/docs/_reference/modules/axon-server-connecto
10	AxonFramework-4.12.1/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.12.1/docs/_reference/modules/axon-server-connecto
11	AxonFramework-4.12.1/docs/_reference/modules/configuration/pages	AxonFramework-4.12.1/docs/_reference/modules/configuratio
12	AxonFramework-4.12.1/docs/_reference/modules/configuration/partials	AxonFramework-4.12.1/docs/_reference/modules/configuratio
13	AxonFramework-4.12.1/docs/_reference/modules/disruptor/pages	AxonFramework-4.12.1/docs/_reference/modules/disrupto
14	AxonFramework-4.12.1/docs/_reference/modules/disruptor/partials	AxonFramework-4.12.1/docs/_reference/modules/disrupto
15	AxonFramework-4.12.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.12.1/docs/_reference/modules/eventsourcin
16	AxonFramework-4.12.1/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.12.1/docs/_reference/modules/eventsourcin
17	AxonFramework-4.12.1/docs/_reference/modules/legacy/pages	AxonFramework-4.12.1/docs/_reference/modules/legac
18	AxonFramework-4.12.1/docs/_reference/modules/legacy/partials	AxonFramework-4.12.1/docs/_reference/modules/legac
19	AxonFramework-4.12.1/docs/_reference/modules/metrics/pages	AxonFramework-4.12.1/docs/_reference/modules/metric
20	AxonFramework-4.12.1/docs/_reference/modules/metrics/partials	AxonFramework-4.12.1/docs/_reference/modules/metric
21	AxonFramework-4.12.1/docs/_reference/modules/micrometer/pages	AxonFramework-4.12.1/docs/_reference/modules/micromete
22	AxonFramework-4.12.1/docs/_reference/modules/micrometer/partials	AxonFramework-4.12.1/docs/_reference/modules/micromete
23	AxonFramework-4.12.1/docs/_reference/modules/modeling/partials	AxonFramework-4.12.1/docs/_reference/modules/modelin
24	AxonFramework-4.12.1/docs/_reference/modules/spring/pages	AxonFramework-4.12.1/docs/_reference/modules/sprin
25	AxonFramework-4.12.1/docs/_reference/modules/spring/partials	AxonFramework-4.12.1/docs/_reference/modules/sprin
26	AxonFramework-4.12.1/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.12.1/docs/_reference/modules/springboot-starte
27	AxonFramework-4.12.1/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.12.1/docs/_reference/modules/springboot-starte
28	AxonFramework-4.12.1/docs/_reference/modules/test/pages	AxonFramework-4.12.1/docs/_reference/modules/tes
29	AxonFramework-4.12.1/docs/_reference/modules/test/partials	AxonFramework-4.12.1/docs/_reference/modules/tes

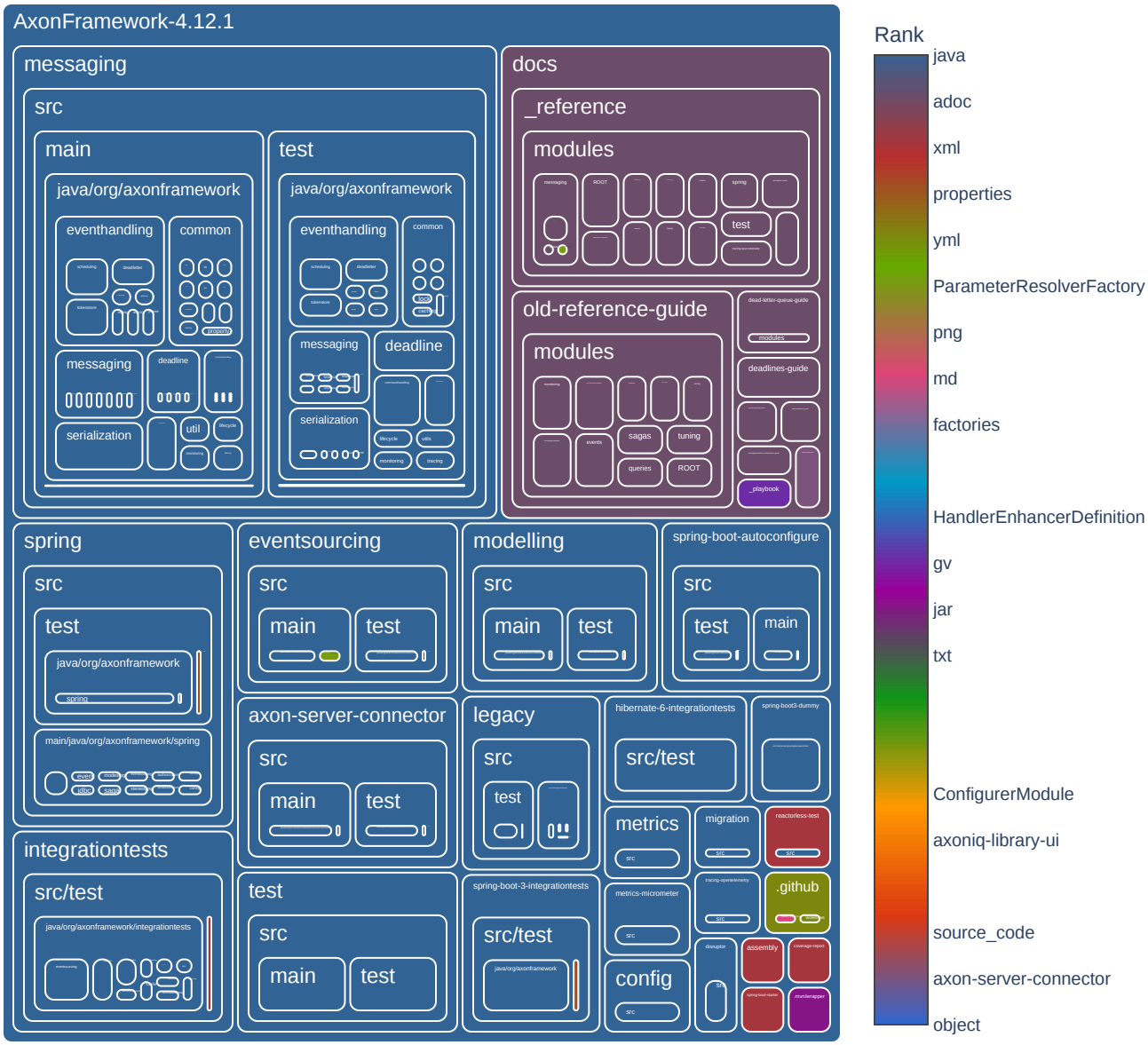
Number of files per directory

Directories and their file count



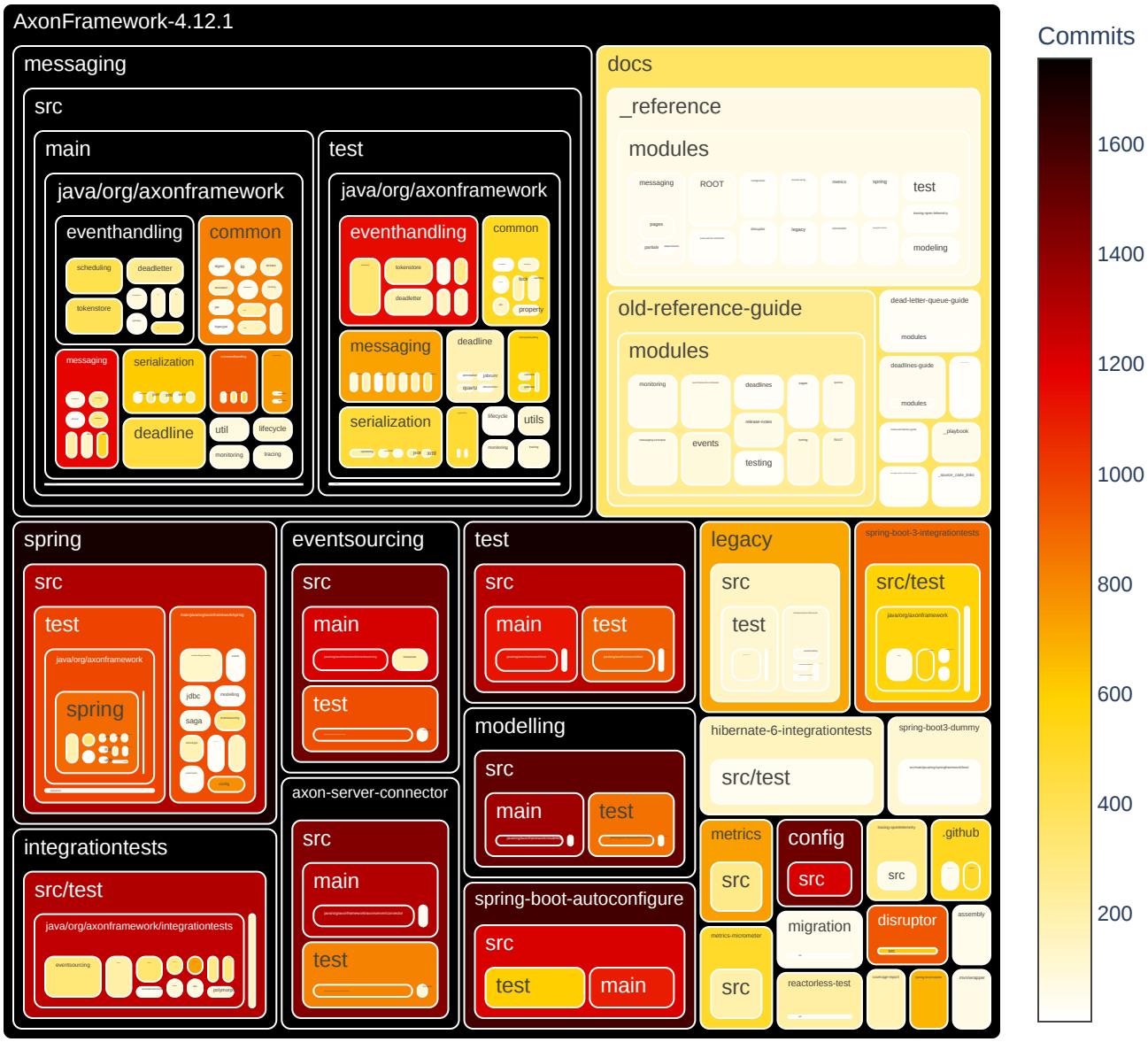
Most frequent file extension per directory

Most frequent file extension per directory



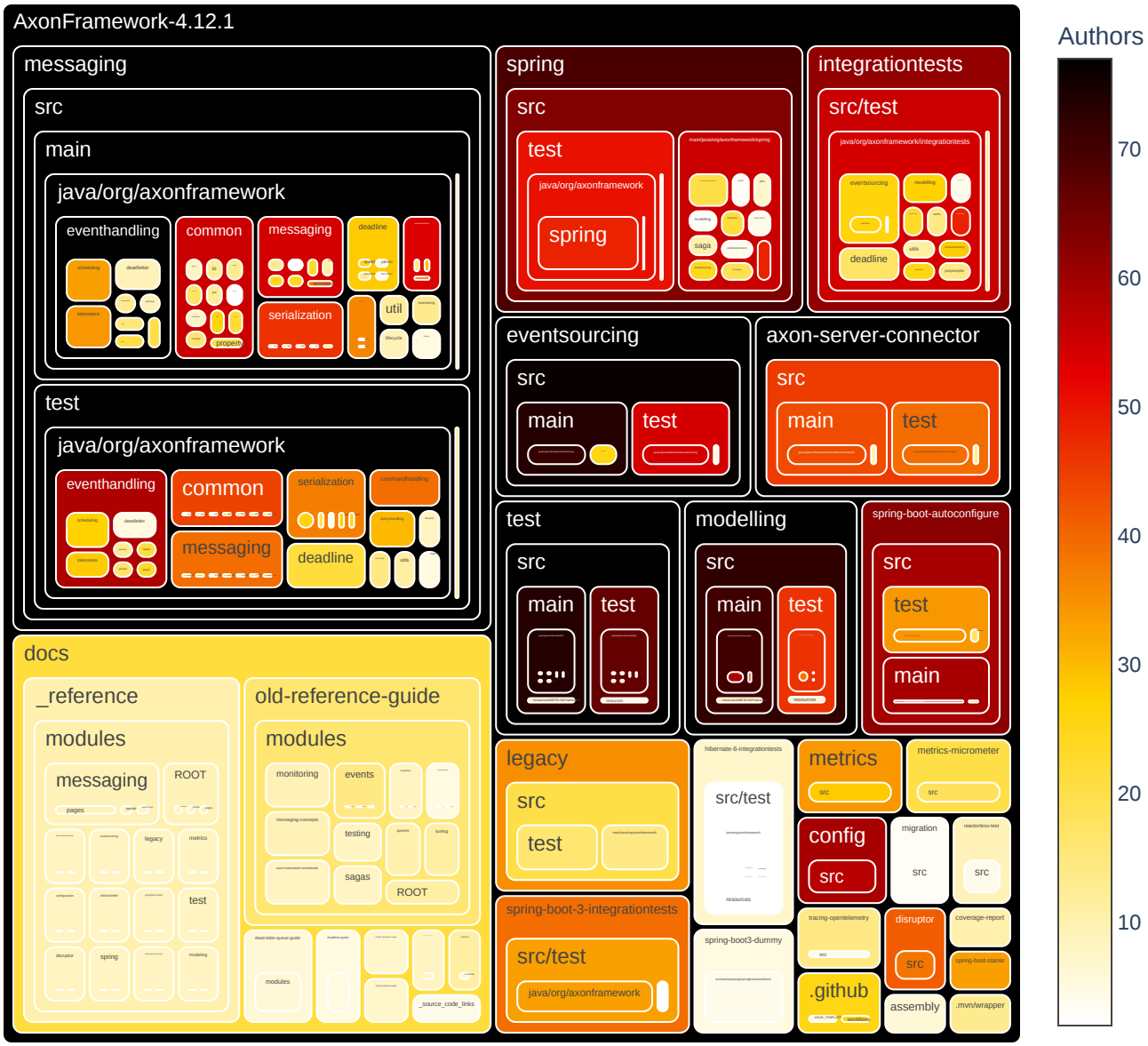
Number of commits per directory

Number of git commits



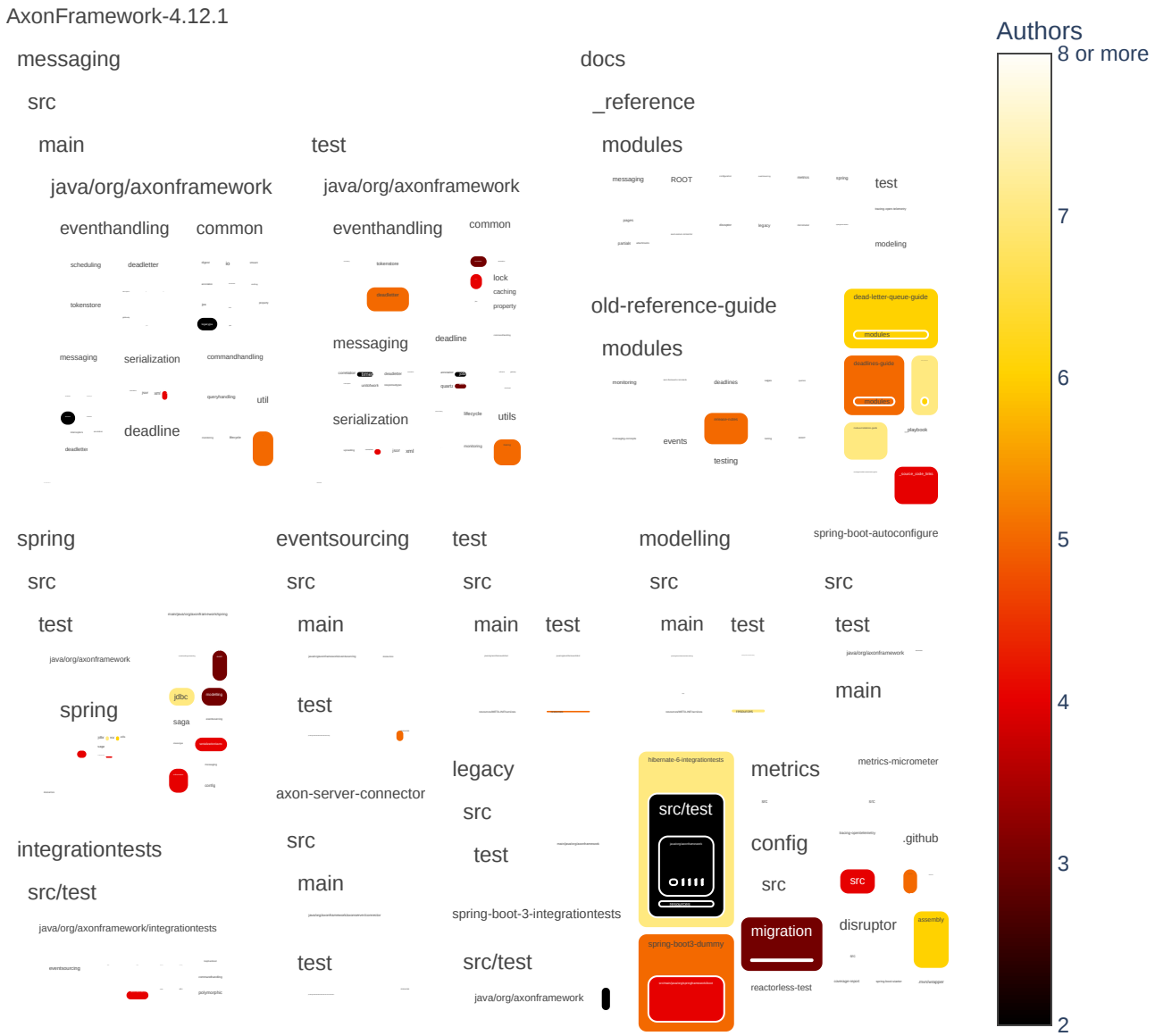
Number of distinct authors per directory

Number of distinct commit authors



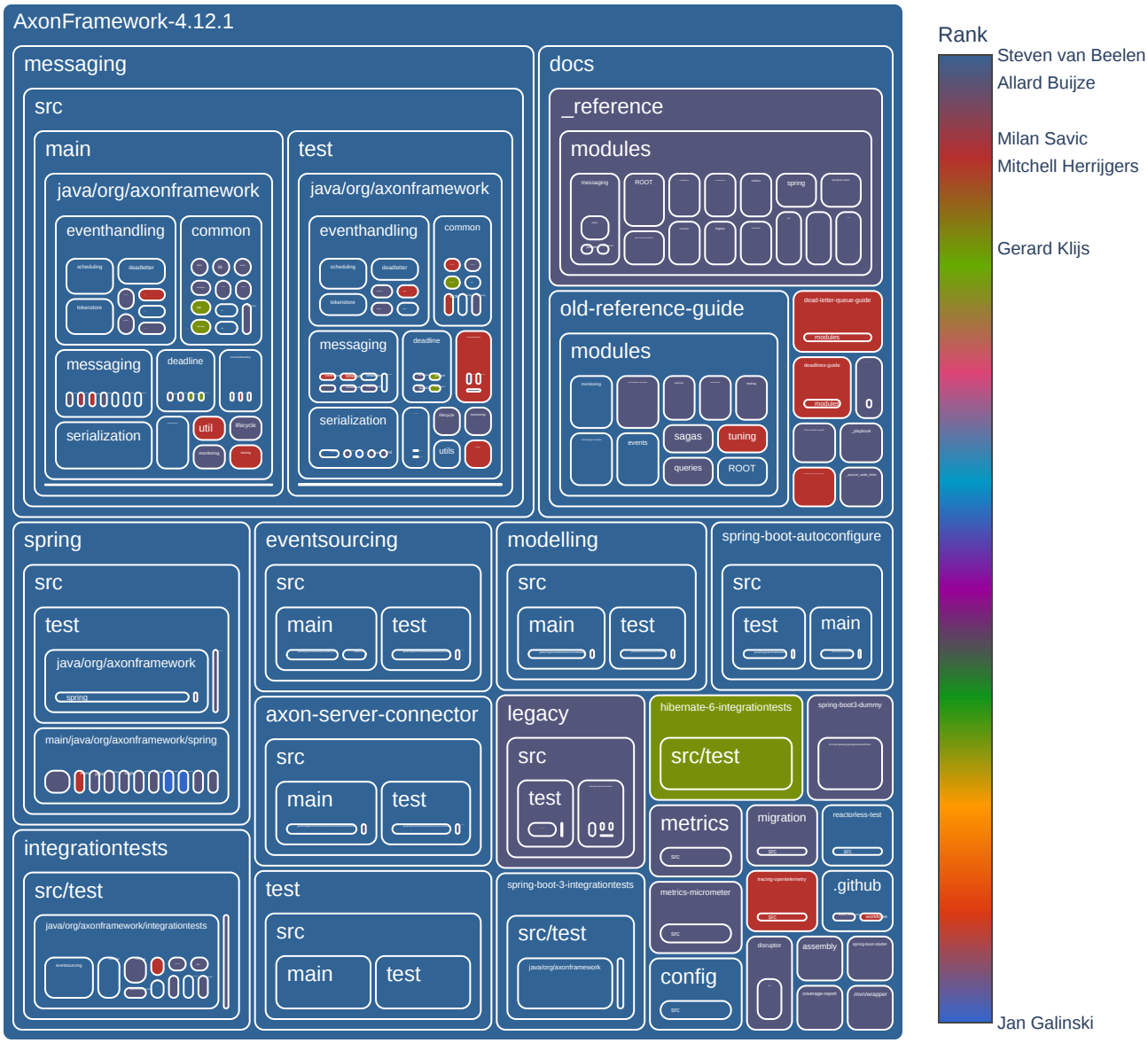
Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)



Main author per directory

Main authors with highest number of commits



Rank

Steven van Beelen
Allard Buijze

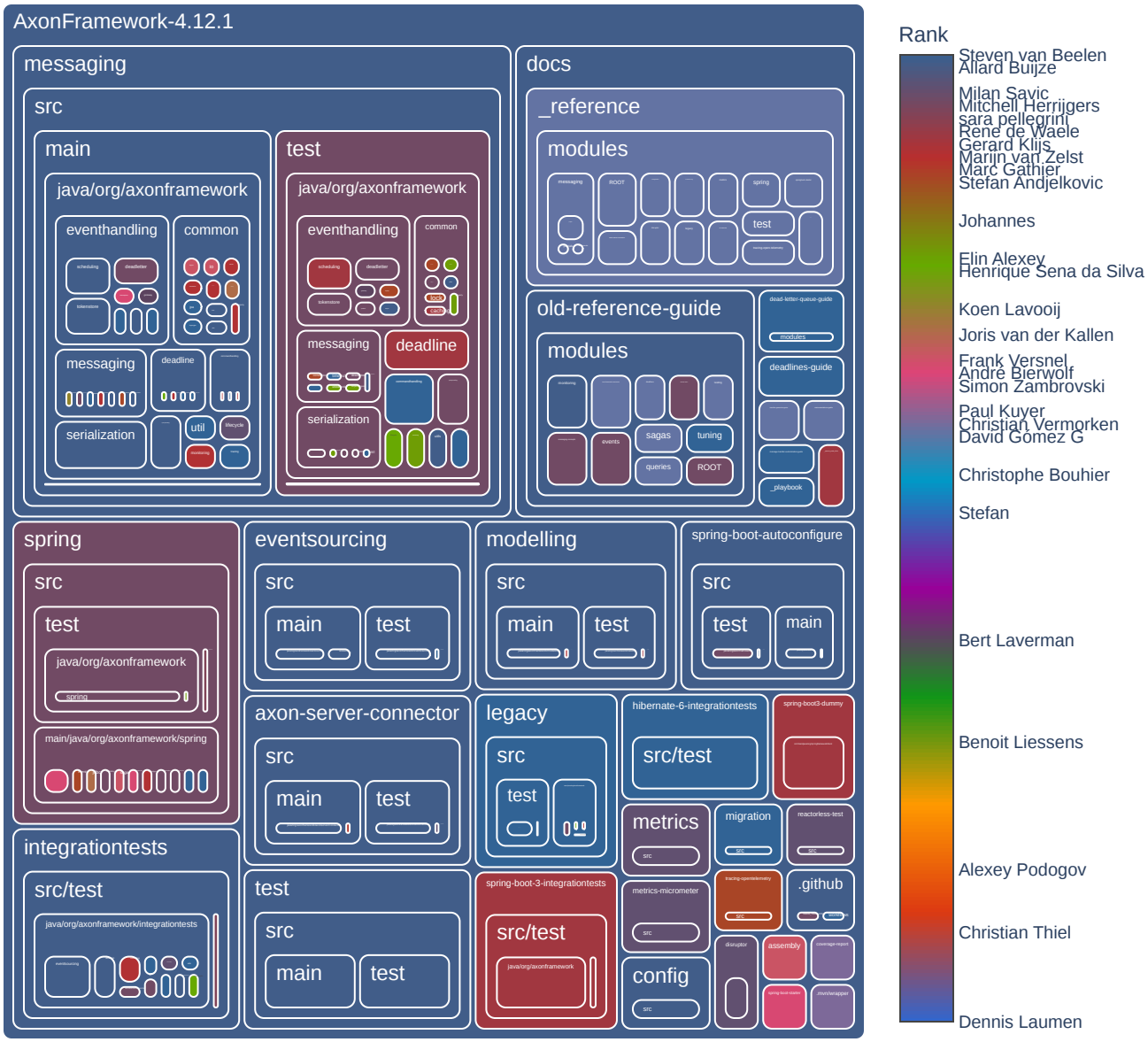
Milan Savic
Mitchell Herrijgers

Gerard Klijs

Jan Galinski

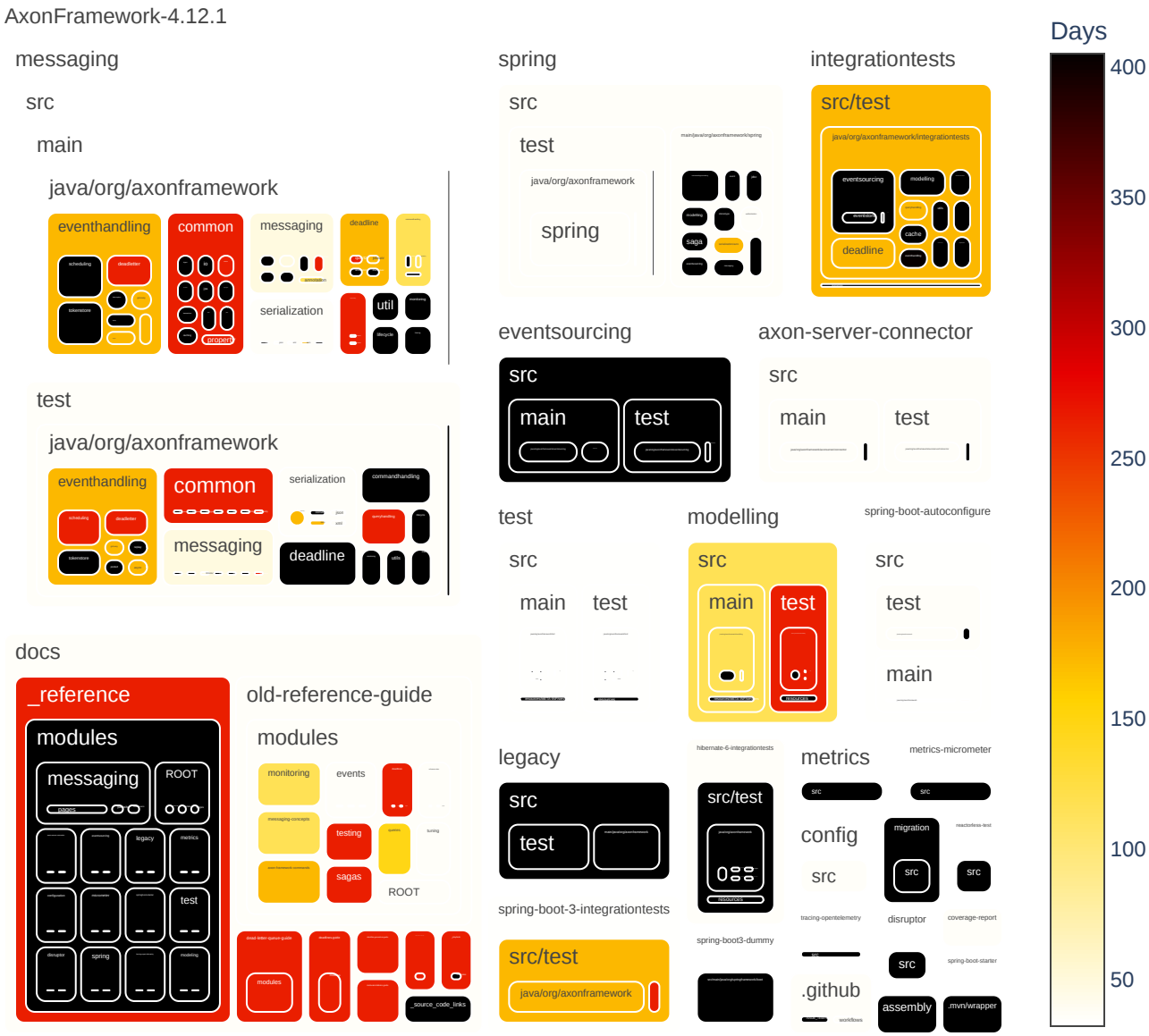
Second author per directory

Second author with the second highest number of commits



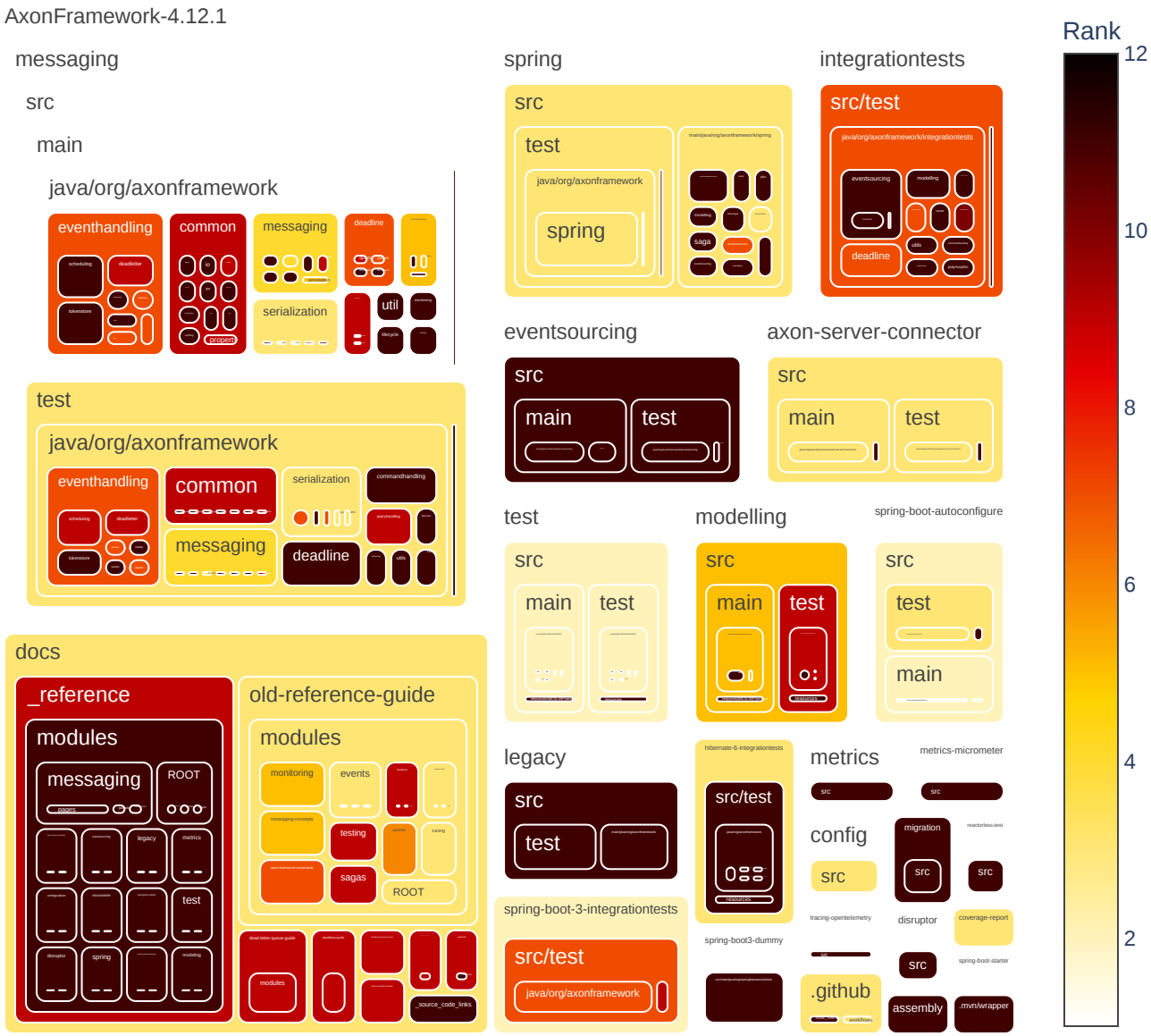
Days since last commit per directory

Days since last commit



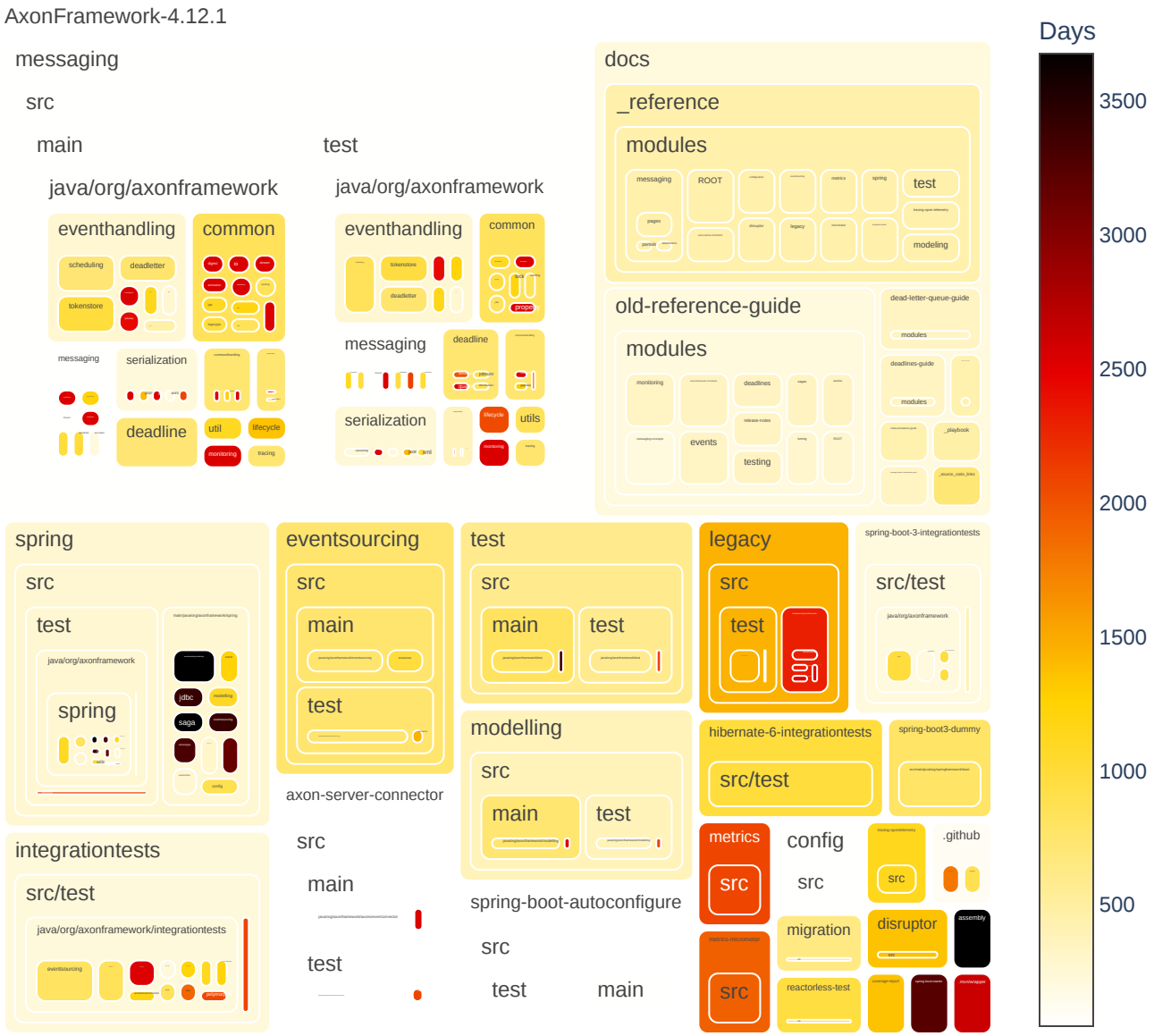
Days since last commit per directory (ranked)

Rank of days since last commit



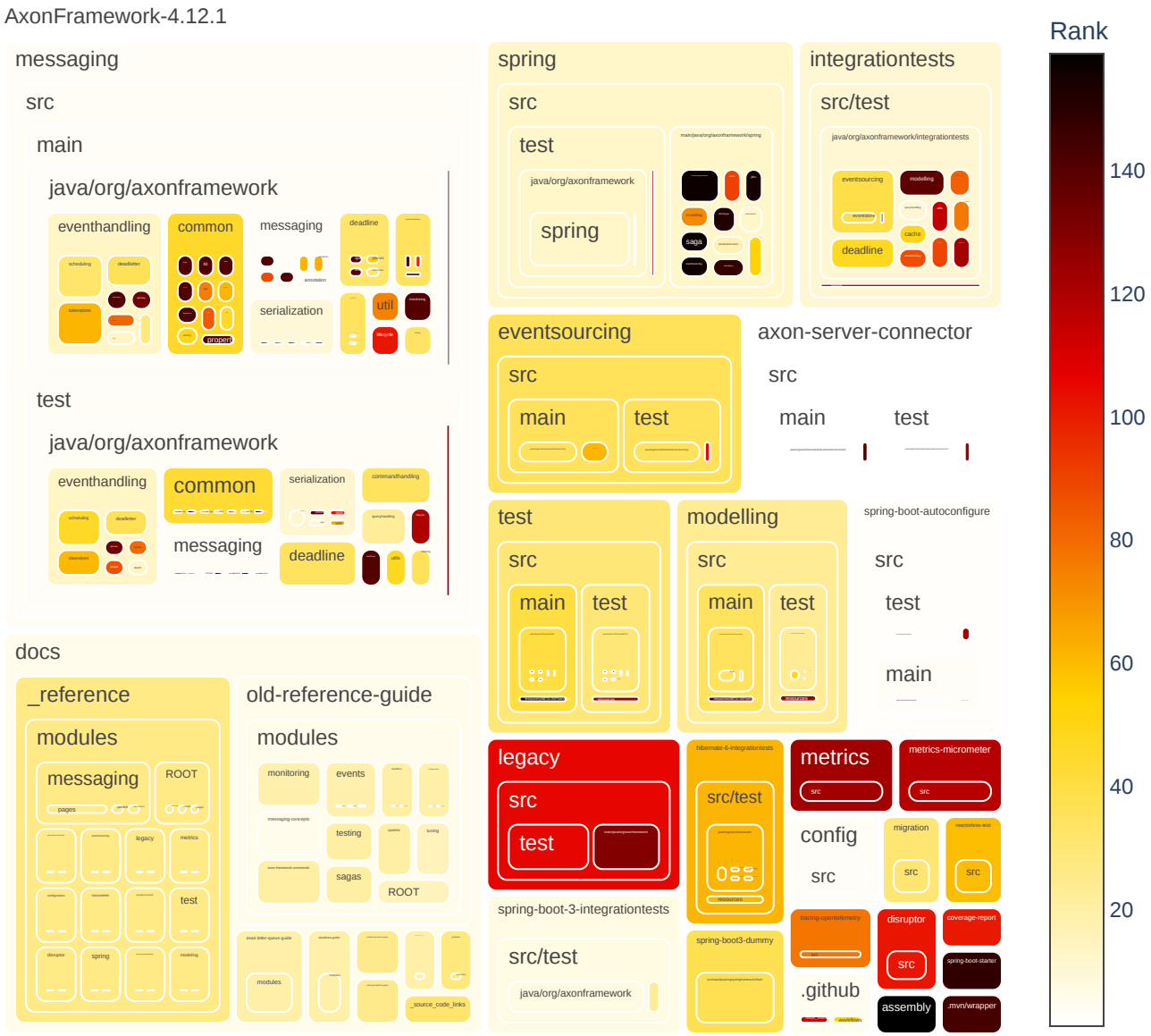
Days since last file creation per directory

Days since last file creation



Days since last file creation per directory (ranked)

Rank of days since last file creation

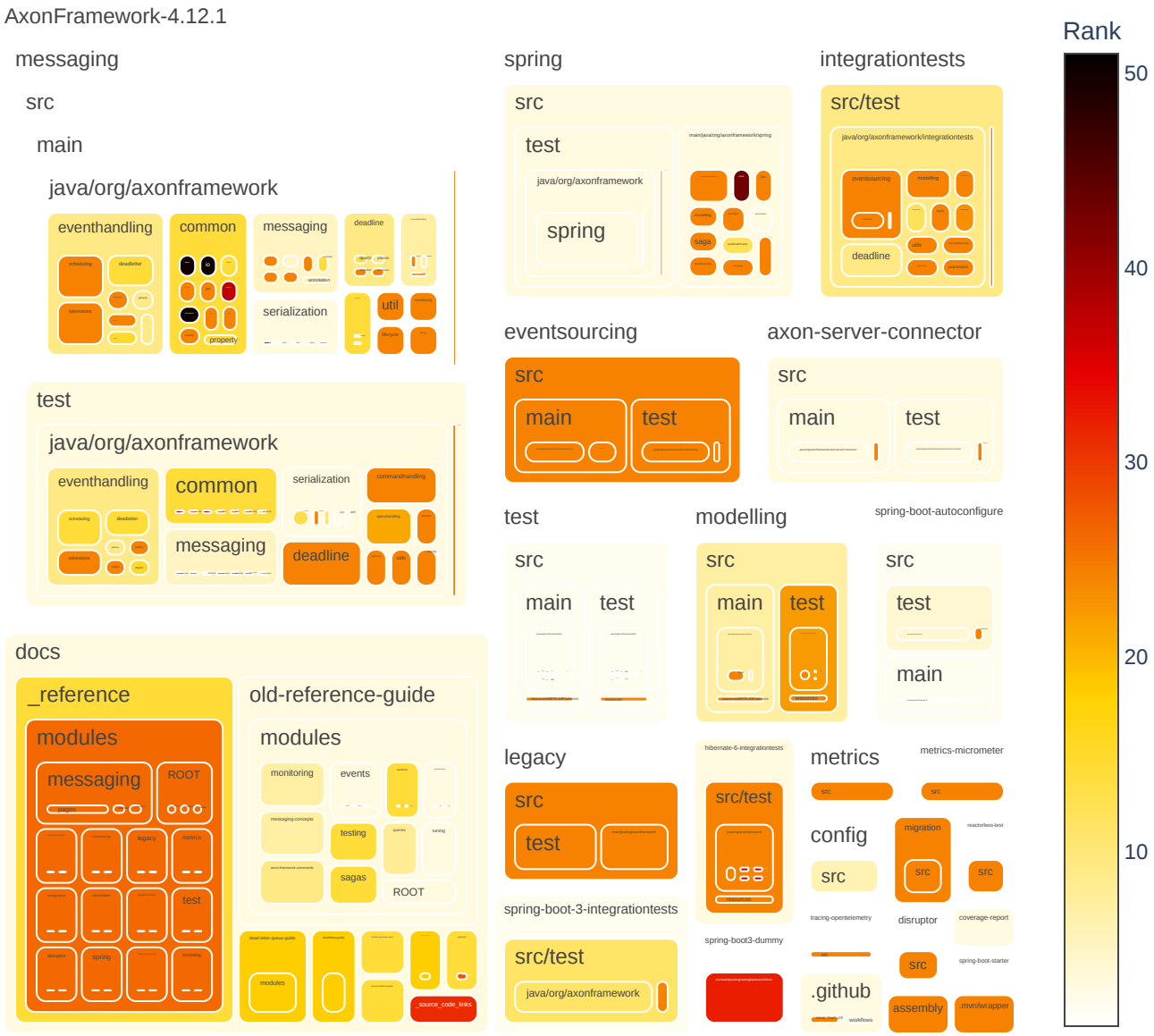


Days since last file modification per directory

Days since last file modification

Days since last file modification per directory (ranked)

Rank of days since last file modification



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

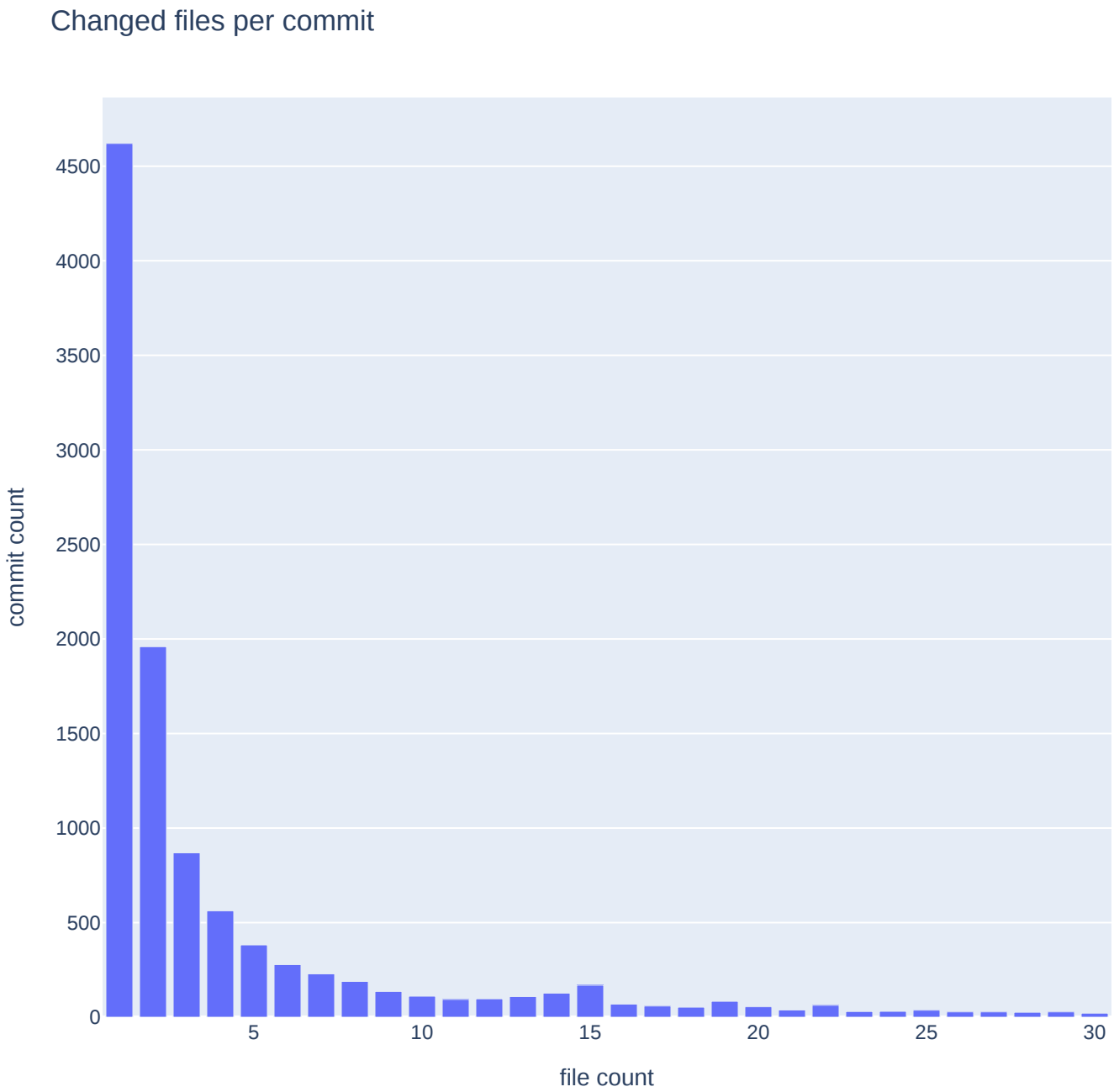
Preview data

```
Sum of commits that changed more than 30 files (each) = 973
Max changed files with one commit = 2178
```

	filesPerCommit	commitCount
count	334.000000	334.000000
mean	343.449102	34.592814
std	439.862566	281.096056
min	1.000000	1.000000
25%	84.250000	1.000000
50%	185.000000	2.000000
75%	426.500000	5.000000
max	2178.000000	4620.000000

	filesPerCommit	commitCount
0	1	4620
1	2	1959
2	3	869
3	4	562
4	5	382
5	6	278
6	7	229
7	8	189
8	9	136
9	10	111
10	11	94
11	12	97
12	13	109
13	14	127
14	15	169
15	16	69
16	17	59
17	18	53
18	19	84
19	20	55
20	21	38
21	22	63
22	23	30
23	24	31
24	25	37
25	26	28
26	27	28
27	28	26
28	29	28
29	30	21

Bar chart with the number of files per commit distribution



Pairwise Changed Files

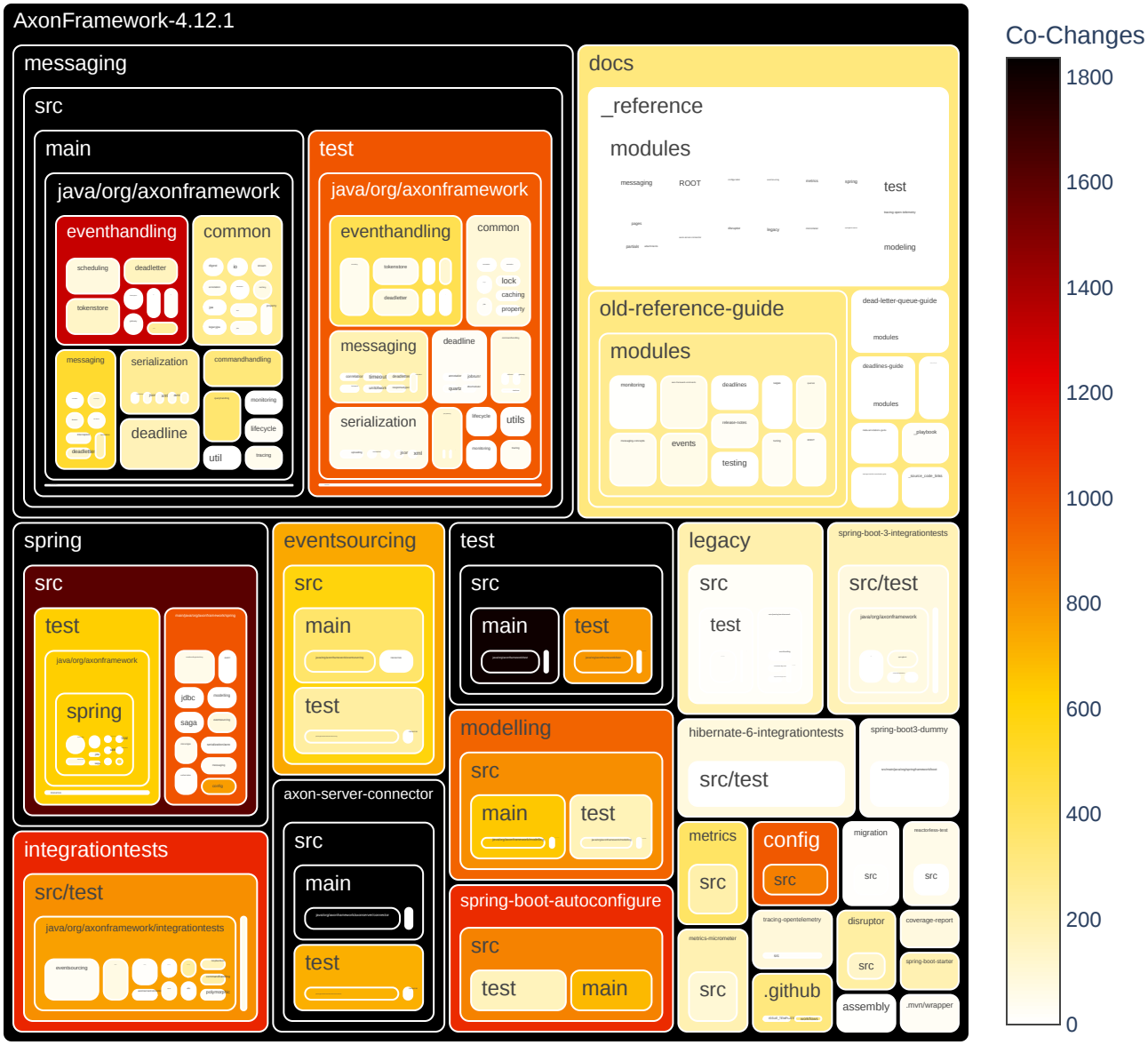
This section analyzes files that where changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count**: The number of commits in which two files were changed together.
- **Commit Lift**: A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity**: The ratio of commits involving either file that also involved both files.

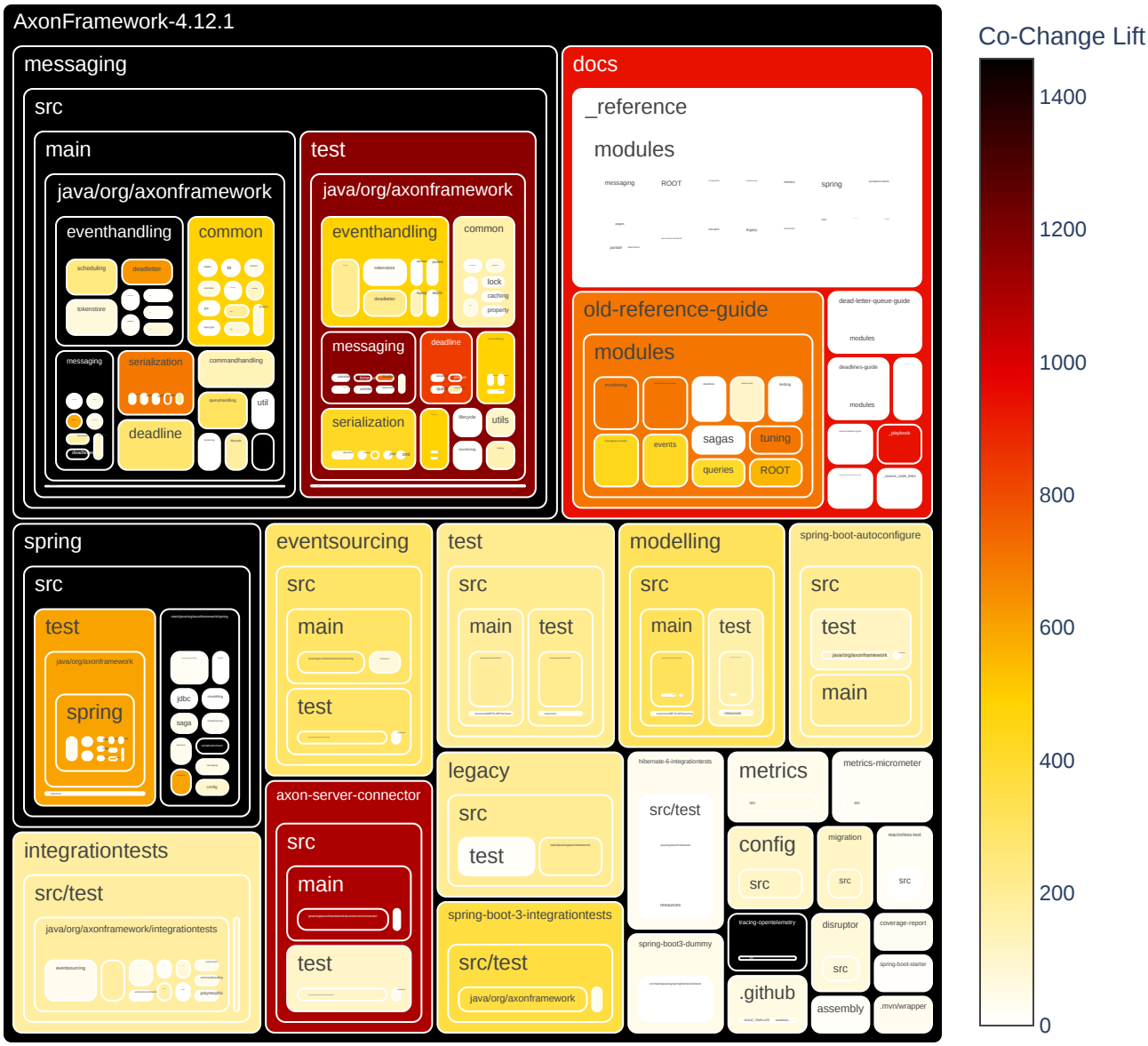
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

Treemap with files changed frequently with others

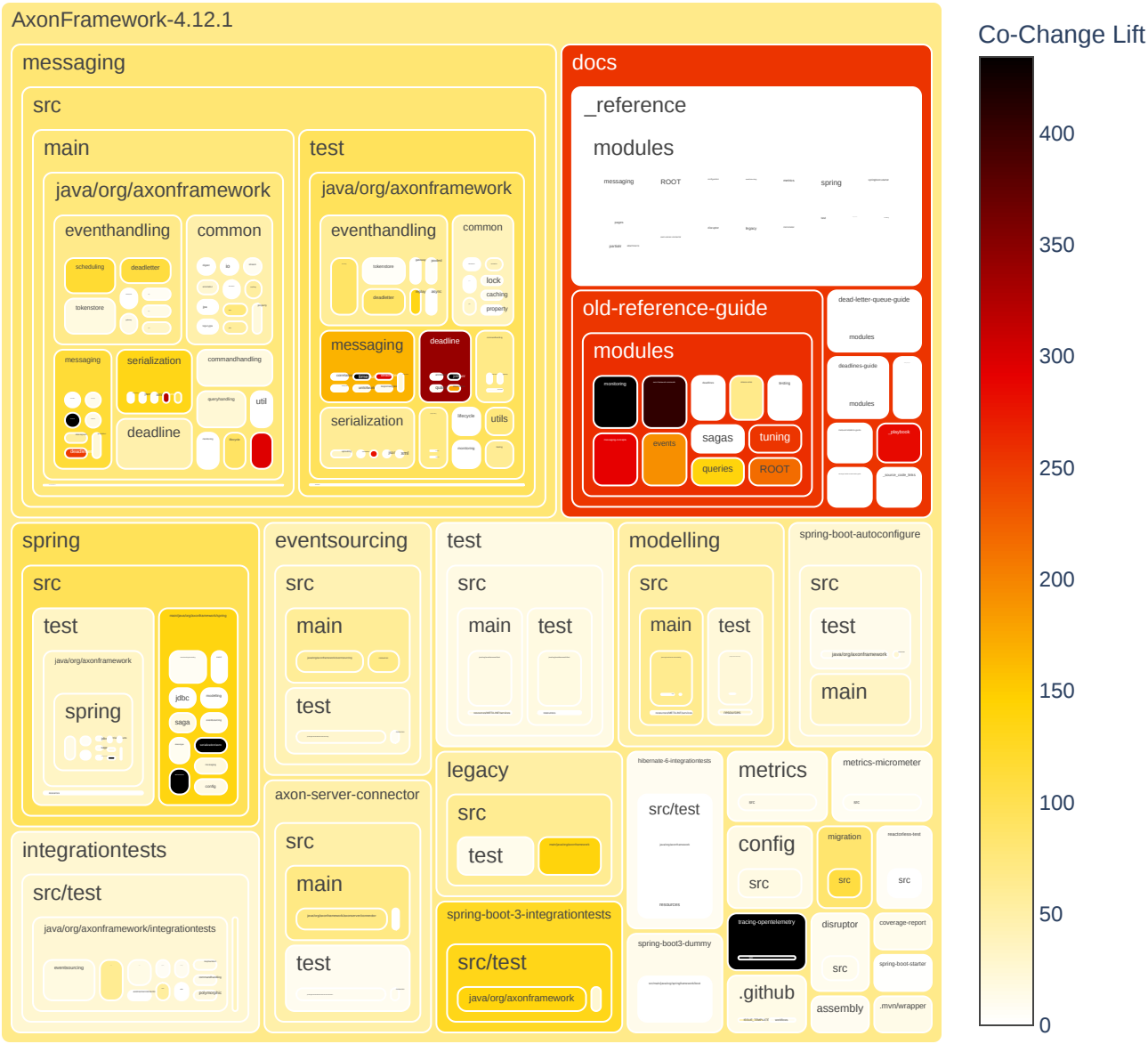
Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

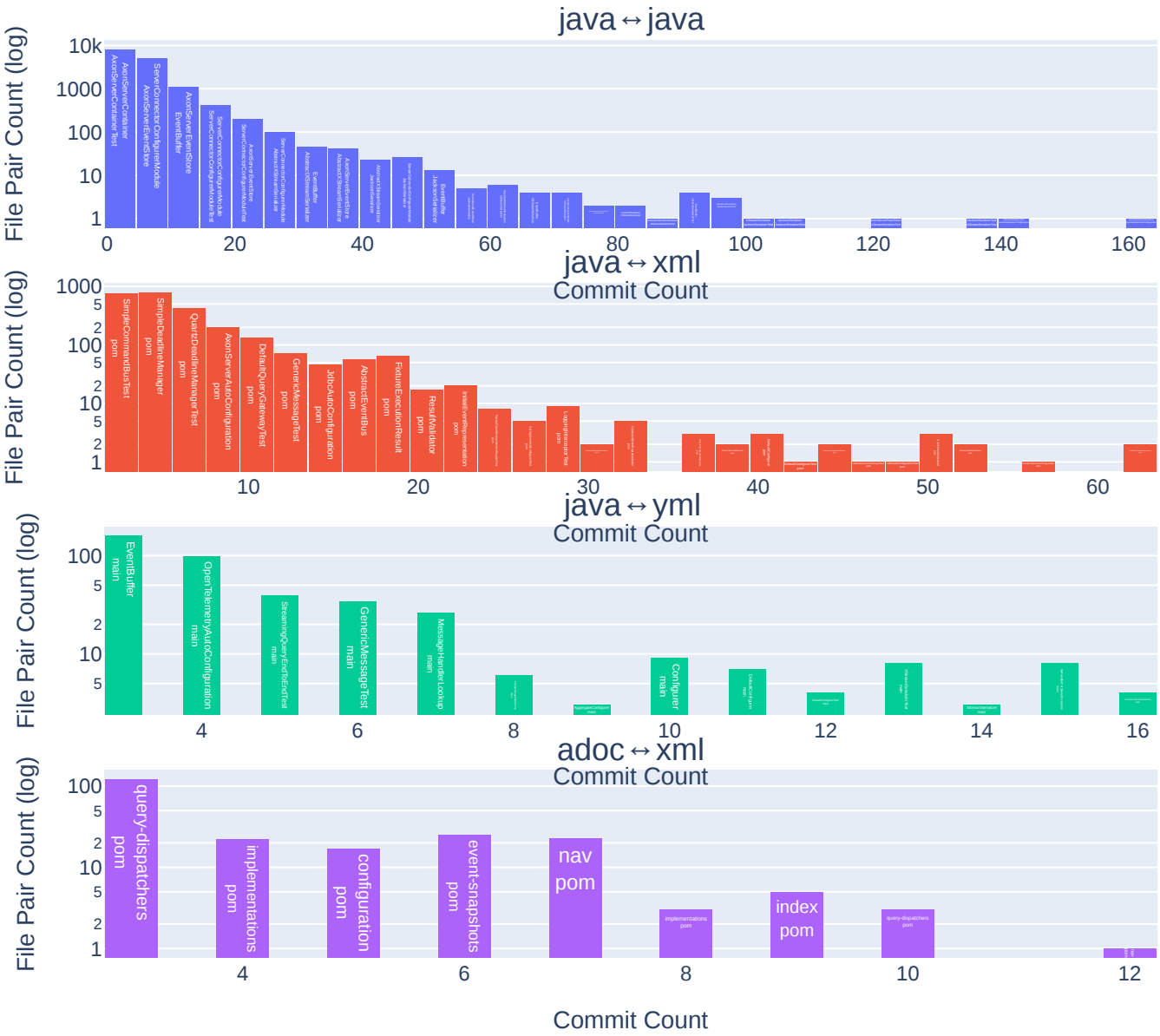


	fileExtensionPair	fileExtensionPairCount
0	java↔java	15189
1	java↔xml	2637
2	java↔yml	408
3	adoc↔xml	221

Files changed together by commit count

	fileExtensionPair	updateCommitCount	GroupRank	filePair	
0	java↔java	163	1	AxonServerQueryBus↔AxonServerCommandBus	
1	java↔java	144	2	AxonServerQueryBus↔AxonServerQueryBusTest	
2	java↔java	135	3	SpringAxonAutoConfigurerTest↔SpringAxonAutoConfigurer	spring/src/tes
3	java↔java	121	4	TrackingEventProcessorTest↔TrackingEventProcessor	integrationtests/src/test/java/org/axonfra
4	java↔java	107	5	SpringAxonAutoConfigurer↔AxonConfiguration	sp
5	java↔java	100	6	EventProcessingModule↔EventProcessingModuleTest	
6	java↔java	96	7	SagaTestFixture↔AggregateTestFixture	
7	java↔java	95	8	AxonServerQueryBusTest↔AxonServerCommandBusTest	
8	java↔java	94	9	AggregateTestFixture↔FixtureConfiguration	
9	java↔java	92	10	AggregateTestFixture↔SpringAxonAutoConfigurer	test/
10	java↔xml	62	1	AggregateTestFixture↔pom	
11	java↔xml	57	2	AggregateTestFixture↔pom	
12	java↔xml	53	3	AggregateTestFixture↔pom	
13	java↔xml	51	4	SpringAxonAutoConfigurer↔pom	
14	java↔xml	48	5	SagaTestFixture↔pom	
15	java↔xml	46	6	SpringAxonAutoConfigurer↔pom	
16	java↔xml	45	7	SpringAxonAutoConfigurerTest↔pom	
17	java↔xml	44	8	SpringAxonAutoConfigurerTest↔pom	
18	java↔xml	42	9	AggregateTestFixture↔pom	
19	java↔xml	41	10	AxonConfiguration↔pom	
20	java↔yml	16	1	JobrunrDeadlineManagerTest↔main	
21	java↔yml	15	2	JobRunrEventSchedulerTest↔main	
22	java↔yml	14	3	TrackingEventProcessor↔main	
23	java↔yml	13	4	SimpleQueryBus↔main	
24	java↔yml	12	5	SimpleEventHandlerInvokerTest↔main	
25	java↔yml	11	6	AxonServerEventStore↔main	
26	java↔yml	10	7	PooledStreamingEventProcessor↔main	
27	java↔yml	9	8	AxonServerQueryBusTest↔main	
28	java↔yml	8	9	DefaultConfigurer↔main	
29	java↔yml	7	10	AggregateAnnotationCommandHandler↔main	
30	adoc↔xml	12	1	minor-releases↔pom	
31	adoc↔xml	10	2	rdbms-tuning↔pom	
32	adoc↔xml	9	3	minor-releases↔pom	
33	adoc↔xml	8	4	minor-releases↔pom	
34	adoc↔xml	7	5	minor-releases↔pom	
35	adoc↔xml	6	6	nav↔pom	
36	adoc↔xml	5	7	event-versioning↔pom	
37	adoc↔xml	4	8	rdbms-tuning↔pom	
38	adoc↔xml	3	9	query-dispatchers↔pom	

Co-Changed Files by their commit count

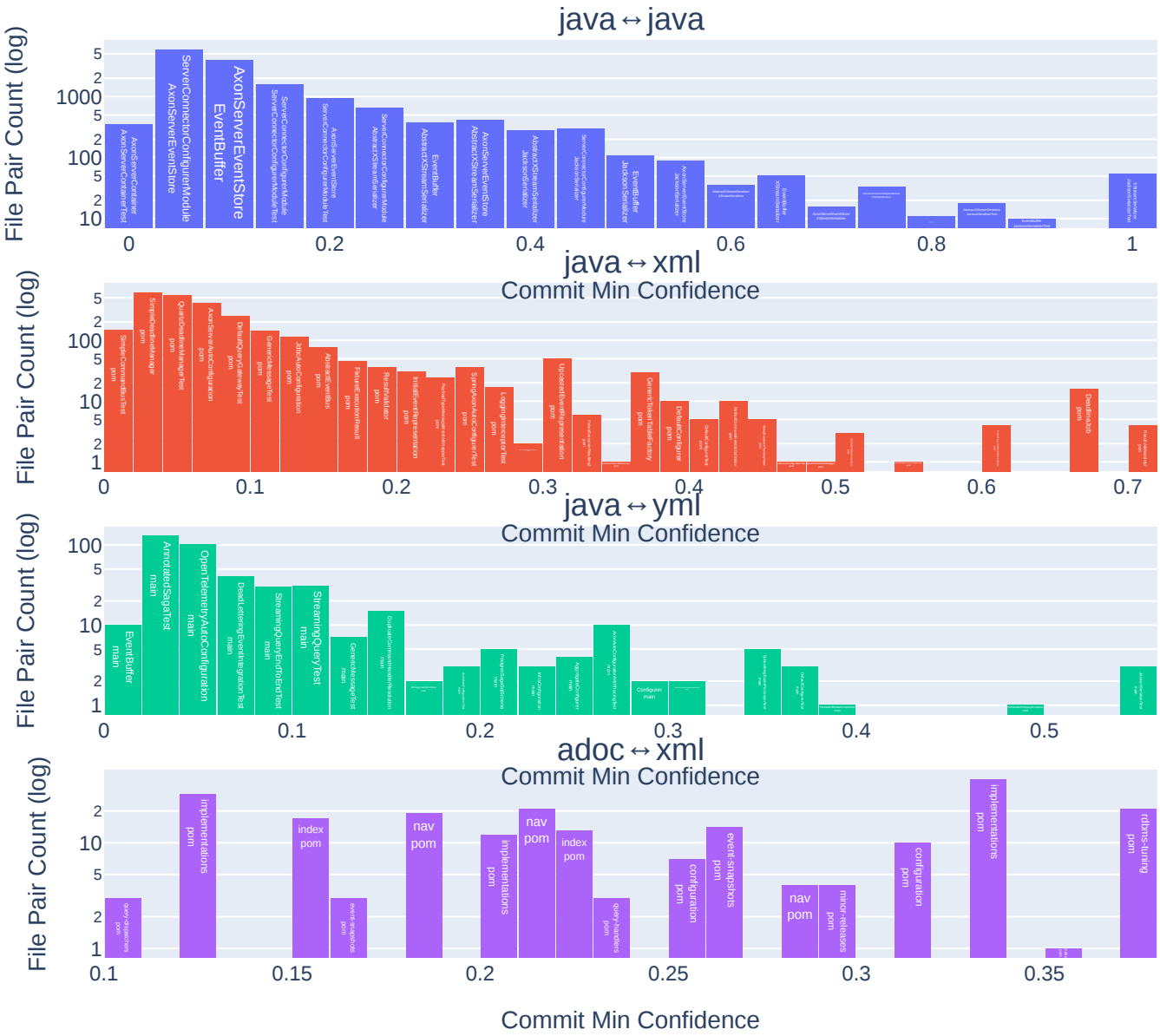


Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

	fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair	
0	java↔java	1.000000	1	AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor	
1	java↔java	0.909091	2	AvroUtil↔SpecificRecordBaseSerializerStrategy	
2	java↔java	0.900000	3	AxonTimeLimitedTask↔TimeoutWrappedMessageHandlingMember	
3	java↔java	0.875000	4	Decisions↔DoNotEnqueue	
4	java↔java	0.857143	5	AvroSerializer↔AvroSerializerStrategy	
5	java↔java	0.833333	6	AvroSerializerStrategy↔ByteArrayToGenericRecordConverter	
6	java↔java	0.800000	7	DeadLetteringEventIntegrationTest↔InMemoryDeadLetteringIntegrationTest	messaging/
7	java↔java	0.785714	8	AvroSerializer↔AvroSerializerTest	
8	java↔java	0.777778	9	JobrunrDeadlineManagerTest↔JobRunrEventSchedulerTest	integrationtes
9	java↔java	0.750000	10	AxonServerAutoConfiguration↔PersistentStreamMessageSource	
10	java↔xml	0.703704	1	JobRunrEventSchedulerTest↔pom	
11	java↔xml	0.666667	2	JobRunrEventSchedulerTest↔pom	
12	java↔xml	0.600000	3	ChainingConverter↔pom	
13	java↔xml	0.545455	4	EventStoreTestUtils↔pom	
14	java↔xml	0.512821	5	DefaultEventMessageConverter↔pom	
15	java↔xml	0.500000	6	ChainingConverter↔pom	
16	java↔xml	0.487179	7	DefaultEventMessageConverter↔pom	
17	java↔xml	0.461538	8	DefaultEventMessageConverter↔pom	
18	java↔xml	0.454545	9	JpaEventStoreAutoConfigurationWithoutAxonServerTest↔pom	
19	java↔xml	0.448276	10	JobrunrDeadlineManagerTest↔pom	
20	java↔yml	0.555556	1	JobRunrEventSchedulerTest↔main	
21	java↔yml	0.481481	2	JobRunrEventSchedulerTest↔antora	
22	java↔yml	0.392857	3	AvroSerializerAutoConfiguration↔antora	
23	java↔yml	0.375000	4	AxonTimeLimitedTaskTest↔main	
24	java↔yml	0.357143	5	AvroSerializerAutoConfiguration↔main	
25	java↔yml	0.340909	6	AbstractDeadlineManagerTestSuite↔main	
26	java↔yml	0.300000	7	ChainingConverter↔main	
27	java↔yml	0.295455	8	AbstractDeadlineManagerTestSuite↔antora	
28	java↔yml	0.288889	9	JobrunrDeadlineManagerTest↔antora	
29	java↔yml	0.277778	10	SpringAxonConfiguration↔main	
30	adoc↔xml	0.375000	1	index↔pom	
31	adoc↔xml	0.352941	2	minor-releases↔pom	
32	adoc↔xml	0.333333	3	nav↔pom	
33	adoc↔xml	0.315789	4	query-dispatchers↔pom	
34	adoc↔xml	0.312500	5	query-handlers↔pom	
35	adoc↔xml	0.291667	6	event-snapshots↔pom	
36	adoc↔xml	0.285714	7	implementations↔pom	
37	adoc↔xml	0.281250	8	rdbms-tuning↔pom	
38	adoc↔xml	0.264706	9	minor-releases↔pom	
39	adoc↔xml	0.263158	10	event-versioning↔pom	

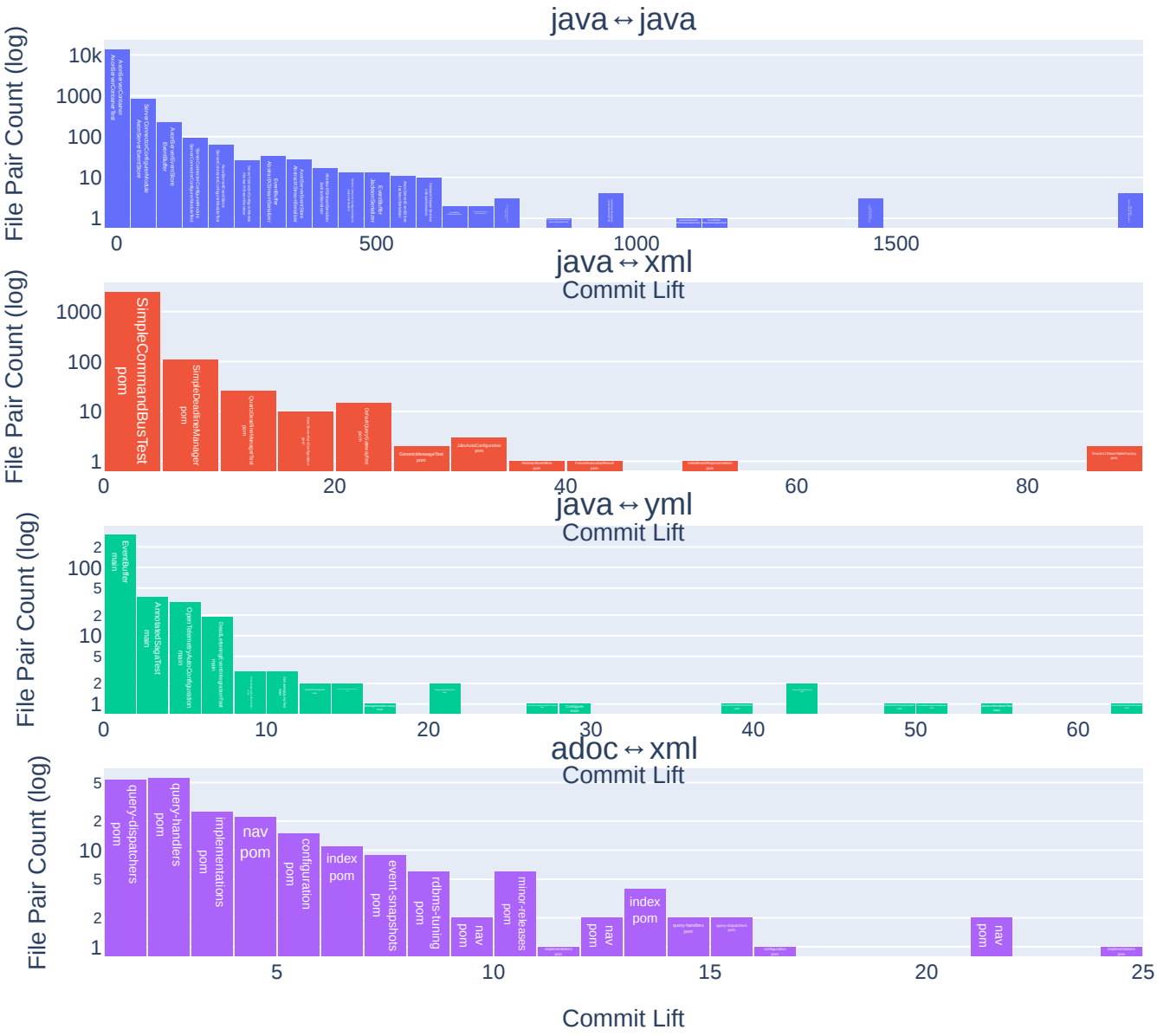
Co-Changed Files by their commit min confidence



Files changed together by commit lift

	fileExtensionPair	updateCommitLift	GroupRank	
0	java↔java	1942.666667	1	AvroSchemaPackages↔Av
1	java↔java	1457.000000	2	DefaultEventProcessorSpanFactory↔DefaultEventProcessorS
2	java↔java	1165.600000	3	TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeout
3	java↔java	1092.750000	4	AxonServerEventStoreFactory↔AxonServerEventS
4	java↔java	971.333333	5	ClasspathAvroSchemaLoader↔SpecificRecordBaseClasspathAvro
5	java↔java	832.571429	6	DeadlineDetailsSerializationTest↔JobRunrDeadlineMan
6	java↔java	728.500000	7	EnqueuePolicy↔NoSuchDeadl
7	java↔java	699.360000	8	AxonServerNonTransientRemoteCommandHandlingException↔AxonServerNonTransientRemoteQueryHan
8	java↔java	693.809524	9	AvroSerializerStrategy↔ByteArrayToGenericR
9	java↔java	647.555556	10	GenericDeadLetterT
10	java↔xml	86.726190	1	ContextAddingInterc
11	java↔xml	50.590278	2	UpstreamAwareStreamObs
12	java↔xml	42.695971	3	JobRunrEventSche
13	java↔xml	35.321212	4	EventStoreTestUt
14	java↔xml	33.996667	5	TestAggregateRoot↔benchmark-
15	java↔xml	33.167480	6	Message↔benchmark-
16	java↔xml	30.118863	7	JobRunrEventSche
17	java↔xml	27.752381	8	MessagingCommandHandler↔benchmark-
18	java↔xml	26.076063	9	JobRunrEventSche
19	java↔xml	24.283333	10	AbstractResourceInject
20	java↔yml	62.357202	1	JobRunrEventSchedu
21	java↔yml	54.637500	2	AbstractResourceInjector↔
22	java↔yml	50.879365	3	AvroSerializerAutoConfig
23	java↔yml	48.566667	4	AxonTimeLimitedT
24	java↔yml	43.753754	5	JobRunrEventSchedulerTest↔slack-rele
25	java↔yml	43.710000	6	ConfigurationScopeAwareProvider↔
26	java↔yml	38.264646	7	AbstractDeadlineManagerTe
27	java↔yml	29.028352	8	JobrunrDeadlineMana
28	java↔yml	26.848894	9	AbstractDeadlineManagerTestSuite↔slack-rele
29	java↔yml	21.479115	10	JpaEventStoreAutoConfigurationWithoutAxonServerTest↔slack-rele
30	adoc↔xml	24.016484	1	
31	adoc↔xml	21.347985	2	
32	adoc↔xml	16.941860	3	
33	adoc↔xml	15.059432	4	conf
34	adoc↔xml	14.667785	5	
35	adoc↔xml	14.231990	6	
36	adoc↔xml	13.723705	7	implem
37	adoc↔xml	13.185520	8	minor
38	adoc↔xml	13.038031	9	conf
39	adoc↔xml	12.008242	10	query

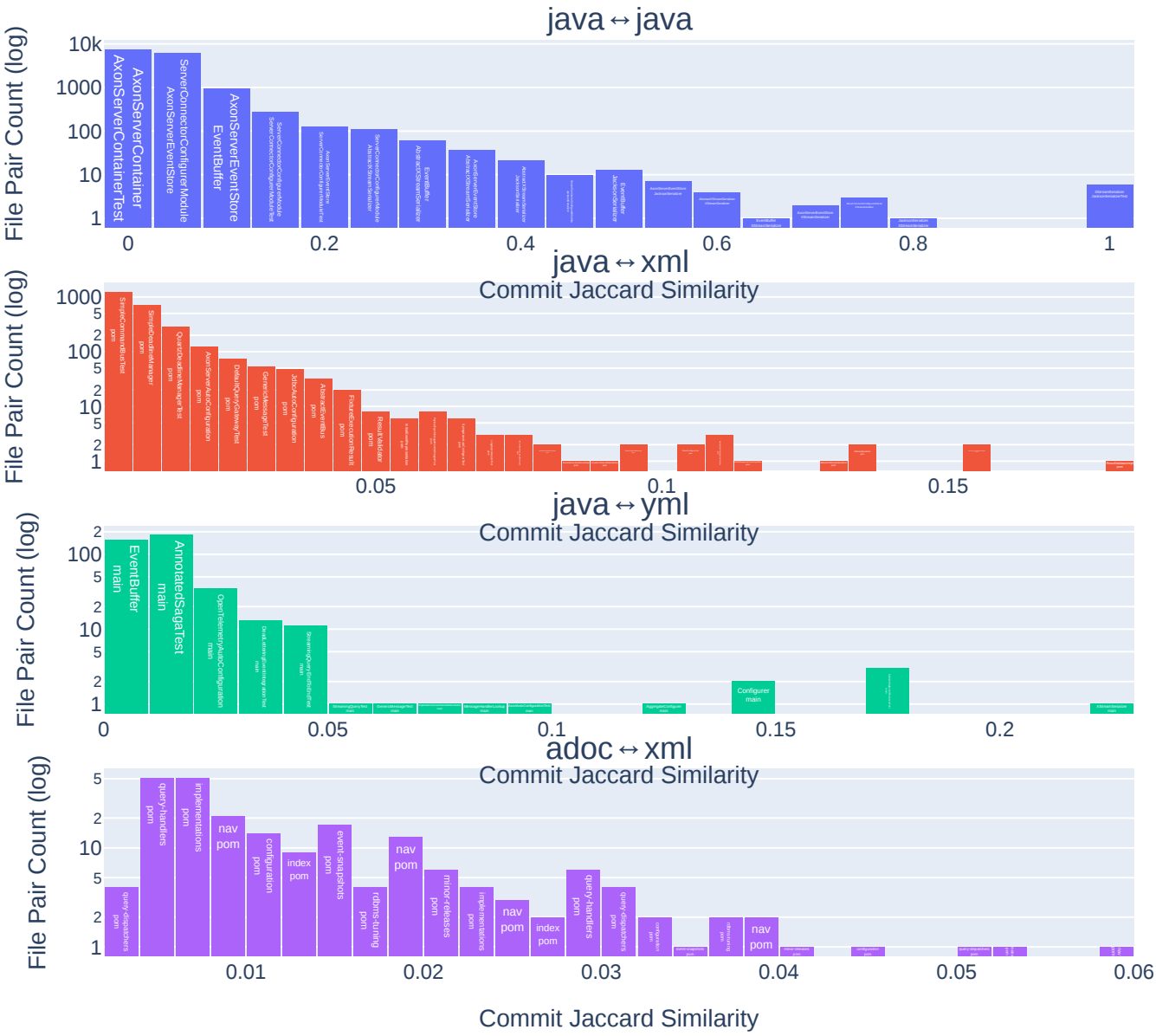
Co-Changed Files by their commit lift



Files changed together by commit Jaccard similarity

	fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	filePair
0	java↔java	1.000000	1	AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor
1	java↔java	0.800000	2	DoNotEnqueueTest↔ShouldEnqueueTest
2	java↔java	0.750000	3	DefaultEventProcessorSpanFactory↔DefaultEventProcessorSpanFactoryTest
3	java↔java	0.714286	4	AvroUtil↔SpecificRecordBaseSerializerStrategy
4	java↔java	0.700000	5	DoNotEnqueue↔ShouldEnqueue
5	java↔java	0.625000	6	AvroSerializerStrategy↔ByteArrayToGenericRecordConverter
6	java↔java	0.600000	7	AxonTimeLimitedTask↔TimeoutWrappedMessageHandlingMember
7	java↔java	0.555556	8	MessageAuthorizationDispatchInterceptor↔MessageAuthorizationDispatchInterceptorTest
8	java↔java	0.545455	9	ShouldEnqueue↔DoNotEnqueueTest
9	java↔java	0.529412	10	AnnotatedChildEntity↔ChildEntity
10	java↔xml	0.180000	1	JobRunrEventSchedulerTest↔pom
11	java↔xml	0.156250	2	ContextAddingInterceptor↔logback
12	java↔xml	0.137405	3	JobrunrDeadlineManagerTest↔pom
13	java↔xml	0.134454	4	AbstractDeadlineManagerTestSuite↔pom
14	java↔xml	0.130435	5	JobRunrEventSchedulerTest↔pom
15	java↔xml	0.113924	6	JobRunrEventSchedulerTest↔pom
16	java↔xml	0.111111	7	TestAggregateRoot↔benchmark-mongo-context
17	java↔xml	0.109375	8	Message↔benchmark-mongo-context
18	java↔xml	0.108974	9	AbstractDeadlineManagerTestSuite↔pom
19	java↔xml	0.106509	10	JobrunrDeadlineManagerTest↔pom
20	java↔yml	0.220339	1	JobRunrEventSchedulerTest↔antora
21	java↔yml	0.177419	2	AvroSerializerAutoConfiguration↔antora
22	java↔yml	0.174419	3	JobRunrEventSchedulerTest↔slack-release-notification
23	java↔yml	0.171053	4	AbstractDeadlineManagerTestSuite↔antora
24	java↔yml	0.145631	5	AbstractDeadlineManagerTestSuite↔slack-release-notification
25	java↔yml	0.144444	6	JobrunrDeadlineManagerTest↔antora
26	java↔yml	0.128205	7	JobrunrDeadlineManagerTest↔slack-release-notification
27	java↔yml	0.090909	8	AbstractResourceInjector↔add-to-project
28	java↔yml	0.081081	9	ConfigurationScopeAwareProvider↔add-to-project
29	java↔yml	0.071429	10	AnnotatedMessageHandlingMemberTest↔antora
30	adoc↔xml	0.059322	1	minor-releases↔pom
31	adoc↔xml	0.053571	2	nav↔pom
32	adoc↔xml	0.051282	3	rdbms-tuning↔pom
33	adoc↔xml	0.044872	4	minor-releases↔pom
34	adoc↔xml	0.040000	5	nav↔pom
35	adoc↔xml	0.039773	6	minor-releases↔pom
36	adoc↔xml	0.038710	7	rdbms-tuning↔pom
37	adoc↔xml	0.036232	8	rdbms-tuning↔pom
38	adoc↔xml	0.036036	9	event-snapshots↔pom
39	adoc↔xml	0.035294	10	nav↔pom

Co-Changed Files by their commit jaccard similarity



Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountExter
0	adoc↔xml	index↔pom	24	
1	adoc↔xml	nav↔pom	27	
2	adoc↔xml	nav↔pom	28	
3	adoc↔xml	configuration↔pom	28	
4	adoc↔xml	minor-releases↔pom	29	
5	adoc↔xml	rdbms-tuning↔pom	35	
6	adoc↔xml	index↔pom	38	
7	adoc↔xml	nav↔pom	39	
8	adoc↔xml	rdbms-tuning↔pom	41	
9	adoc↔xml	minor-releases↔pom	41	
10	java↔java	TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoutInterceptorTest	83	
11	java↔java	AvroSchemaPackages↔AvroSchemaScan	83	
12	java↔java	AvroSchemaScan↔ClasspathAvroSchemaLoader	83	
13	java↔java	AvroSchemaPackages↔ClasspathAvroSchemaLoader	83	
14	java↔java	MetadataContextGetter↔MetadataContextSetter	83	
15	java↔java	DefaultEventProcessorSpanFactory↔DefaultEventProcessorSpanFactoryTest	86	
16	java↔java	NoSuchDeadLetterException↔IgnoreTest	86	
17	java↔java	SpanUtils↔SpanUtilsTest	86	
18	java↔java	DoNotEnqueueTest↔ShouldEnqueueTest	92	
19	java↔java	AvroSerializerStrategy↔ByteArrayToGenericRecordConverter	99	
20	java↔xml	JobRunrEventSchedulerTest↔pom	36	
21	java↔xml	JobRunrEventSchedulerTest↔pom	44	
22	java↔xml	JobRunrEventSchedulerTest↔pom	47	
23	java↔xml	JobRunrEventSchedulerTest↔pom	62	
24	java↔xml	AbstractDeadlineManagerTestSuite↔pom	72	
25	java↔xml	ContextAddingInterceptor↔logback	74	
26	java↔xml	TokenAddingInterceptor↔logback	74	
27	java↔xml	JobRunrEventSchedulerTest↔pom	80	
28	java↔xml	JobrunrDeadlineManagerTest↔pom	84	
29	java↔xml	UpstreamAwareStreamObserver↔logback	84	
30	java↔yml	JobRunrEventSchedulerTest↔antora	8	
31	java↔yml	JobRunrEventSchedulerTest↔slack-release-notification	11	
32	java↔yml	AvroSerializerAutoConfiguration↔antora	14	
33	java↔yml	AbstractDeadlineManagerTestSuite↔slack-release-notification	22	
34	java↔yml	AbstractDeadlineManagerTestSuite↔antora	23	
35	java↔yml	JobrunrDeadlineManagerTest↔antora	27	
36	java↔yml	AxonTimeLimitedTaskTest↔antora	33	
37	java↔yml	JobrunrDeadlineManagerTest↔slack-release-notification	33	
38	java↔yml	JobRunrEventSchedulerTest↔pullrequest	34	
39	java↔yml	JobRunrEventSchedulerTest↔main	36	

Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

	filePair	combinedMetricsScore	updateCommitCount	updateCommitMinCon
0	AxonTimeLimitedTask↔UnitOfWorkTimeoutInterceptor	100	14	1
1	TimeoutWrappedMessageHandlingMemberTest↔UnitOfWorkTimeoutInterceptorTest	83	5	1
2	EventProcessingModule↔EventProcessingModuleTest	2786	100	0
3	AvroSchemaPackages↔AvroSchemaScan	83	3	1
4	AvroSchemaScan↔ClasspathAvroSchemaLoader	83	3	1
5	AvroSchemaPackages↔ClasspathAvroSchemaLoader	83	3	1
6	JobrunrDeadlineManagerTest↔JobRunrEventSchedulerTest	340	21	0
7	JobRunrEventSchedulerTest↔AbstractDeadlineManagerTestSuite	345	17	0
8	AxonServerQueryBus↔AxonServerCommandBus	3563	163	0
9	SimpleEventHandlerInvoker↔SimpleEventHandlerInvokerTest	888	43	0
10	DeadLetteringEventHandlerInvoker↔DeadLetteringEventHandlerInvokerTest	627	22	0
11	JpaAutoConfiguration↔JpaJavaxAutoConfiguration	736	22	0
12				